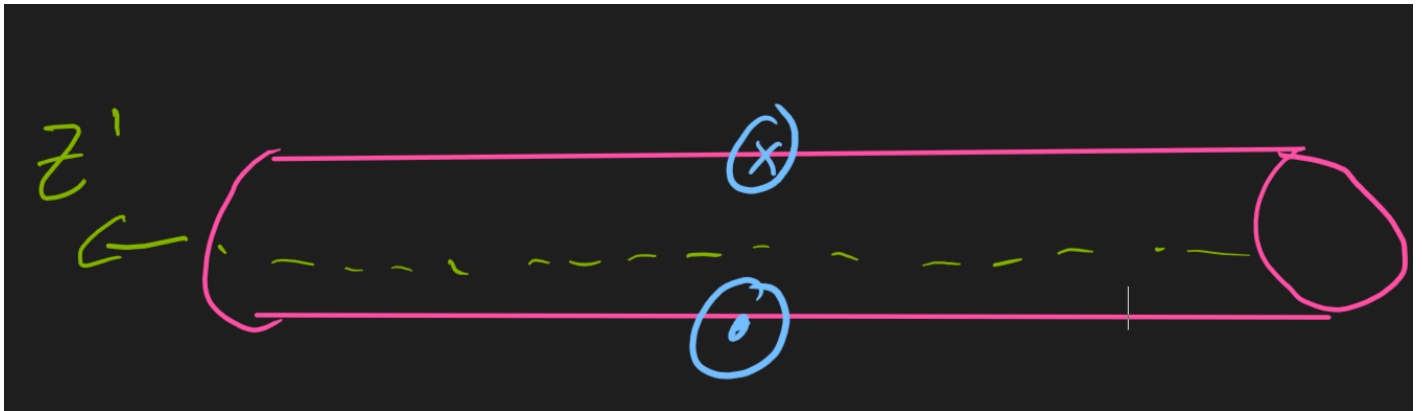


Wk 12 HW

Physics 111: Theoretical Mechanics Problem Set # 9

1. (10 points) Accelerating Space Station. A cylindrically symmetric space station (principal axis of symmetry $\hat{z}'$, and $\lambda_1 = \lambda_2 \equiv \lambda$) is floating in free space. It has rockets mounted symmetrically on either side that are firing and exert a constant net torque Γ oriented along $\hat{z}'$.

Solve Euler's equations exactly for $\vec{\omega}$ in the body frame, and describe the motion. As an initial condition, take $\vec{\omega} = \omega_{10}\hat{x}' + \omega_{30}\hat{z}'$ at $t = 0$. *Hint: in class, we solved the coupled equations by differentiating twice. Here, we use another, more useful trick in the case of time-dependent $\omega_{z'}$: define $\eta = \omega_{x'} + i\omega_{y'}$, and combine the differential equations for $\omega_{x'}$ and $\omega_{y'}$ into a single differential equation for $\eta = \dot{\omega}_{x'} + i\dot{\omega}_{y'}$. Solve the equation for η , then take the real and imaginary parts of η to find solutions for $\omega_{x'}$ and $\omega_{y'}$.*



Let the blue arrows denote rocket boosters to rotate the rocket barrel.

We know that

$$\left(\frac{d\vec{L}}{dt} \right)_{\text{inertial}} = \left(\frac{d\vec{L}}{dt} \right)_{\text{rot}} + \vec{\omega} \times \vec{L}$$

$$\vec{L} = I\vec{\omega}$$

$$\vec{\Gamma} = I\vec{\alpha}$$

Because the rockets are exerting torque on the craft, we have

$$\frac{d\vec{L}}{dt} = \vec{\Gamma}$$

Which we can plug in to get

$$\vec{\Gamma} = \left(\frac{d\vec{L}}{dt} \right)_{\text{rot}} + \vec{\omega} \times \vec{L}_{\text{inertial}}$$

We also know that in the rotating frame

$$\vec{L} = \lambda_1 \omega_{x'} \hat{x}' + \lambda_2 \omega_{y'} \hat{y}' + \lambda_3 \omega_{z'} \hat{z}'$$

so we can calculate $\vec{\omega} \times \vec{L}$ as

$$\vec{\omega} \times \vec{L} = (\omega_{x'} \hat{x}' + \omega_{y'} \hat{y}' + \omega_{z'} \hat{z}') \times (\lambda_1 \omega_{x'} \hat{x}' + \lambda_2 \omega_{y'} \hat{y}' + \lambda_3 \omega_{z'} \hat{z}')$$

which tells us

$$\left(\frac{d\vec{L}}{dt} \right)_{rot} = \lambda_1 \dot{\omega}_{x'} \hat{x}' + \lambda_2 \dot{\omega}_{y'} \hat{y}' + \lambda_3 \dot{\omega}_{z'} \hat{z}'$$

Using $\lambda_1 = \lambda_2$, we can write out the Euler equations with Torque

$$\begin{aligned} \lambda \dot{\omega}'_x &= (\lambda - \lambda_3) \omega'_y \omega'_z + \Gamma_x \\ \lambda \dot{\omega}'_y &= (\lambda_3 - \lambda) \omega'_x \omega'_z + \Gamma_y \\ \lambda_3 \dot{\omega}'_z &= (\lambda - \lambda) \omega'_x \omega'_y + \Gamma_z \end{aligned}$$

Γ is only oriented along the z direction, so

$$\begin{aligned} \lambda \dot{\omega}'_x &= (\lambda - \lambda_3) \omega'_y \omega'_z \\ \lambda \dot{\omega}'_y &= (\lambda_3 - \lambda) \omega'_x \omega'_z \\ \lambda_3 \dot{\omega}'_z &= \Gamma_z \end{aligned}$$

So $\dot{\omega}_{z'} = \frac{\Gamma_z}{\lambda_3}$.

We can easily solve that part as

$$\omega_{z'} = \frac{z}{\lambda_3} t$$

Lets solve for $\dot{\omega}_{x'}$ and $\dot{\omega}_{y'}$ toget ω_x and ω_y :

$$\begin{aligned} \lambda \dot{\omega}_{x'} &= (\lambda - \lambda_3) \omega_{y'} \frac{\Gamma_z}{\lambda_3} \\ \lambda \dot{\omega}_{y'} &= (\lambda_3 - \lambda) \omega_{x'} \frac{\Gamma_z}{\lambda_3} \end{aligned}$$

So we have finally

$$\dot{\vec{\omega}} = \frac{(\lambda - \lambda_3) \omega_{y'}}{\lambda \lambda_3} \Gamma_z$$

$$\dot{\vec{\omega}} = \frac{(\lambda_3 - \lambda) \omega_{x'}}{\lambda \lambda_3} \Gamma_z$$

$$\frac{\Gamma_z}{\lambda_3}$$

These are basically constants, so I'll rewrite this with

$$= \frac{(\lambda - \lambda_3)}{\lambda \lambda_3}$$

$$\omega_{y'} \Gamma_z$$

$$\dot{\vec{\omega}} = -\omega_{x'} \Gamma_z$$

$$\frac{\Gamma_z}{\lambda_3}$$

Note for the future, the relation

$$\omega_{y'} = \frac{\dot{\omega}_{x'}}{\Gamma_z}$$

We can simplify this with expressions that we do have (knowing that the torque is constant):

$$\vec{\omega} = \begin{pmatrix} \dot{\omega}_{y'} \Gamma_z \\ -\dot{\omega}_{x'} \Gamma_z \\ 0 \end{pmatrix}$$

We can replace expressions that we already have circularly

$$\vec{\omega} = \begin{pmatrix} \omega_{x'} \Gamma_z^2 \\ -\omega_{y'} \Gamma_z^2 \\ 0 \end{pmatrix}$$

Lets take the ansatz that our final differential equation will have the form

$$\vec{\omega}(t) = \vec{\omega}_0 e^{i\omega t}$$

Where $\omega_{x'} = \omega_0 \cos(\Gamma_z t)$, $\omega_{y'} = \omega_0 \sin(\Gamma_z t)$,

We have the initial conditions for $\vec{\omega}_0$

$$\vec{\omega}_0 = \begin{pmatrix} \omega_{10} \hat{x}' \\ \omega_{20} \hat{y}' \\ 0 \end{pmatrix}$$

For the $\hat{x}'$ component, we have

$$\omega_{x'} = \omega_{10} \cos(\Gamma_z t)$$

We can also solve for $\omega_{y'}$ with the expression from earlier, $\omega_{y'} = \frac{\dot{\omega}_{x'}}{\Gamma_z}$

$$\frac{d}{dt} \omega_{x'} = -\Gamma_z \sin(\Gamma_z t)$$

So

$$\omega_{y'} = \omega_{20} \sin(\Gamma_z t)$$

2. (16 points) Chandler Wobble.

- (a) (3 points) Imagine that the Earth is a perfectly rigid, uniform, and spherical body of radius R spinning around its axis with angular velocity $\vec{\omega}$. Does the Earth experience any precession? Explain why or why not.
- (b) (8 points) By an act of God, a mountain of mass $m \ll M_{\text{Earth}}$ appears at an angle α from the rotation axis. The angular momentum is unchanged. The Earth subsequently undergoes free precession as described in class. Identify a set of principal axes, calculate the components of the inertia tensor in your principal axis basis, and determine the body-frame precession frequency, Ω_b . Write your answer in terms of m , M_{Earth} , ω , and α . *Hint: The diagonal components of the inertia tensor of a sphere are all equal to $2/5MR^2$.*
- (c) (5 points) The North Pole is defined as the point on the surface of the Earth lying in the direction of $\vec{\omega}$. How long would it take the North Pole to move a distance $d = 100$ miles from its current position? Compute the numerical value using $m = 10^{-8} M_{\text{Earth}}$, $R = 4000$ miles, $\alpha = 60^\circ$, and $\omega = 2\pi \text{ day}^{-1}$.

3. (24 points) Rolling Cone. A solid, uniform cone of mass M , height h , and radius R lies on a flat table. It rolls without slipping along the flat table as illustrated in the figure; essentially, the cone rolls in a large circle while the tip remains at the origin. The polar axis is taken to be perpendicular to the table ($\hat{z}$). The goal is to find the kinetic energy of the cone.

- (a) (4 points) Find the location of the centre of mass of the cone.
- (b) (8 points) Describe the motion of the centre of mass, and show that the kinetic energy of the motion of the centre of mass as a function of $\dot{\phi}$ (where ϕ is the azimuthal angle of the axis of the cone) is

$$T_{\text{CM}} = \frac{9Mh^4}{32(h^2 + R^2)}\dot{\phi}^2. \quad (1)$$

- (c) (8 points) Using the standard set of Euler angles (θ, ϕ, ψ) that describe the orientation of the body relative to its centre of mass, argue that θ is fixed and express θ in terms of R and h . Using the rolling without slipping condition, derive a relationship between ψ and ϕ .
- (d) (4 points) Using our expression for the kinetic energy of a rigid body in terms of Euler's angles, show that the rotational kinetic energy about the CM is

$$T_{\text{rot}} = \frac{1}{2}\lambda\dot{\phi}^2\frac{h^2}{h^2 + R^2} + \frac{1}{2}\lambda_3\dot{\phi}^2\frac{h^4}{R^2(h^2 + R^2)}. \quad (2)$$

You can leave your answer in terms of the principal moments λ and λ_3 ; you do not need to find λ and λ_3 (but can if you want to!).

- (e) (optional, ungraded) Using the method of Lagrange multipliers, find the force of friction (magnitude and direction) to keep the cone in this motion. *Hint: think carefully about the net force acting on the cone.*

